R Project Practice

Load our data

We start with loading our data into the variable "clev". This variable will be used throughout the notebook to access our data.

```
clev = read.csv("ClevelandHeart.csv")
```

A reminder of the column names and descriptions:
1. Age
2. Sex
3. Chestpain
4. RestBP (resting blood pressure, mm Hg)
5. Chol (cholesterol, mg/dl)
6. Fbs (fasting blood sugar > 120 mg/dl, TRUE or FALSE)
7. RestECG (resting electrocardiographic results, 0=normal, 1=ST-T wave abnormality, 2=left ventricular hypertrophy)
8. MaxHR (maximum heart rate achieved)
9. ExAng (exercise induced angina, yes or no)
10. Fluoroscopy (number of major vessels (0-3) colored by fluoroscopy)
11. Thal (Thalassemia is an inherited blood disorder)
12. AHD (patient diagnosed with heart disease yes or no)

Basics

$

Access our variable's columns with "$". We can view the fasting blood sugar (T/F) using the following:

```
clev$Fbs
```

```
##   [1]  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
##  [13]  TRUE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
##  [25] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE
##  [37] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE
##  [49]  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [61] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE
##  [73] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE
##  [85] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [97] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE
## [109] FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE FALSE  TRUE FALSE  TRUE FALSE
## [121]  TRUE FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE
## [133] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE
## [145] FALSE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE
## [157] FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE
## [169] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE
## [181] FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE
## [193]  TRUE  TRUE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE
```

```
## [205] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
## [217] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [229] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE
## [241] FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
## [253] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
## [265] FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [277] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE
## [289] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE
## [301] FALSE FALSE FALSE
```

== (equal to)

We can create a TRUE/FALSE vector by comparing using "==". Here we're comparing if the type of chest pain our samples have is "typical":

```
clev$ChestPain == "typical"
```

```
##   [1]  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE FALSE FALSE
##  [25] FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE
##  [37] FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE
##  [49] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE
##  [61] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [73] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [85] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [97] FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE
## [109] FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [121] FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [133] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
## [145] FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE
## [157] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [169] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [181] FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [193] FALSE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE
## [205] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE  TRUE
## [217] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [229] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [241] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [253] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE
## [265] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE
## [277] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [289] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE
## [301] FALSE FALSE FALSE
```

Note: 2 equal signs is comparison: "==", but 1 equal sign is assignment.

One outputs T/F whether an individual is labeled "male" and one changes all rows for "Sex" to male! We can count how many males there are after each command to verify. Comparing:

```
clev$Sex == "male"
```

```
##   [1]  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE FALSE  TRUE  TRUE  TRUE FALSE
##  [13]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
##  [25]  TRUE FALSE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [37]  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE  TRUE  TRUE
##  [49] FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
```

```
##  [61] FALSE FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE
##  [73]  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE
##  [85]  TRUE  TRUE  TRUE FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE FALSE  TRUE
##  [97]  TRUE FALSE  TRUE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE
## [109]  TRUE  TRUE FALSE  TRUE  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## [121]  TRUE FALSE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## [133]  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [145]  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE FALSE  TRUE  TRUE  TRUE
## [157]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE FALSE
## [169]  TRUE FALSE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [181]  TRUE FALSE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [193]  TRUE FALSE FALSE  TRUE  TRUE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE
## [205]  TRUE  TRUE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## [217] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE
## [229]  TRUE  TRUE FALSE FALSE  TRUE FALSE FALSE  TRUE  TRUE  TRUE FALSE  TRUE
## [241]  TRUE FALSE FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [253]  TRUE FALSE  TRUE FALSE FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE  TRUE
## [265]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE
## [277] FALSE FALSE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE  TRUE
## [289]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## [301]  TRUE FALSE  TRUE
```

```
sum(clev$Sex == "male")
```

```
## [1] 206
```

Assigning:

```
clev$Sex = "male"
sum(clev$Sex == "male")
```

```
## [1] 303
```

Even if you correct this mistake and add another "=", the data will need to be imported again!

We will reload our data so the rest of the notebook will run as expected:

```
clev = read.csv("ClevelandHeart.csv")
```

Subsetting data

Data can be subset with "[]", e.g. variablename[row,col]
clev[,y] selects col y in our data (across all rows) clev[x,] selects row x in our data (across all columns)

```
clev[1,]
```

```
##   Age  Sex ChestPain RestBP Chol  Fbs RestECG MaxHR ExAng Fluoroscopy  Thal AHD
## 1  63 male   typical    145  233 TRUE       2   150    no           0 fixed  No
```

columns can be selected by column index (a number) or by name:

```
clev[,2]
```

```
##   [1] "male"   "male"   "male"   "male"   "female" "male"   "female" "female"
##   [9] "male"   "male"   "male"   "female" "male"   "male"   "male"   "male"
##  [17] "male"   "male"   "female" "male"   "male"   "female" "male"   "male"
##  [25] "male"   "female" "female" "female" "male"   "male"   "female" "male"
##  [33] "male"   "male"   "male"   "male"   "male"   "male"   "male"   "male"
##  [41] "female" "male"   "female" "male"   "female" "male"   "male"   "male"
##  [49] "female" "male"   "female" "male"   "male"   "male"   "male"   "male"
```

```
##  [57] "male"   "male"   "male"   "male"   "female" "female" "male"   "female"
##  [65] "male"   "male"   "male"   "male"   "male"   "male"   "female" "male"
##  [73] "male"   "male"   "male"   "female" "male"   "female" "male"   "male"
##  [81] "male"   "female" "male"   "male"   "male"   "male"   "male"   "female"
##  [89] "female" "female" "male"   "female" "male"   "female" "female" "male"
##  [97] "male"   "female" "male"   "male"   "male"   "male"   "female" "female"
## [105] "male"   "male"   "male"   "male"   "male"   "male"   "female" "male"
## [113] "male"   "female" "female" "male"   "male"   "female" "male"   "male"
## [121] "male"   "female" "male"   "male"   "male"   "female" "female" "male"
## [129] "male"   "female" "male"   "male"   "male"   "male"   "female" "female"
## [137] "male"   "male"   "male"   "male"   "male"   "male"   "male"   "male"
## [145] "male"   "male"   "male"   "male"   "male"   "female" "male"   "female"
## [153] "female" "male"   "male"   "male"   "male"   "male"   "male"   "male"
## [161] "male"   "male"   "female" "female" "male"   "male"   "male"   "female"
## [169] "male"   "female" "male"   "male"   "female" "female" "male"   "male"
## [177] "male"   "male"   "male"   "male"   "male"   "female" "male"   "male"
## [185] "female" "female" "male"   "male"   "male"   "male"   "male"   "male"
## [193] "male"   "female" "female" "male"   "male"   "female" "female" "male"
## [201] "female" "female" "male"   "female" "male"   "male"   "male"   "male"
## [209] "male"   "female" "female" "male"   "male"   "female" "male"   "male"
## [217] "female" "female" "female" "male"   "female" "female" "female" "male"
## [225] "female" "female" "male"   "female" "male"   "male"   "female" "female"
## [233] "male"   "female" "female" "male"   "male"   "male"   "female" "male"
## [241] "male"   "female" "female" "male"   "female" "male"   "male"   "male"
## [249] "male"   "male"   "male"   "male"   "male"   "female" "male"   "female"
## [257] "female" "female" "male"   "male"   "female" "female" "female" "male"
## [265] "male"   "male"   "male"   "male"   "male"   "male"   "male"   "male"
## [273] "male"   "female" "male"   "male"   "female" "female" "male"   "female"
## [281] "male"   "male"   "female" "male"   "male"   "male"   "female" "male"
## [289] "male"   "male"   "male"   "female" "male"   "male"   "female" "male"
## [297] "male"   "female" "male"   "male"   "male"   "female" "male"
```

has the same output as:

```
clev[,"Sex"]
```

```
##   [1] "male"   "male"   "male"   "male"   "female" "male"   "female" "female"
##   [9] "male"   "male"   "male"   "female" "male"   "male"   "male"   "male"
##  [17] "male"   "male"   "female" "male"   "male"   "female" "male"   "male"
##  [25] "male"   "female" "female" "female" "male"   "male"   "female" "male"
##  [33] "male"   "male"   "male"   "male"   "male"   "male"   "male"   "male"
##  [41] "female" "male"   "female" "male"   "female" "male"   "male"   "male"
##  [49] "female" "male"   "female" "male"   "male"   "male"   "male"   "male"
##  [57] "male"   "male"   "male"   "male"   "female" "female" "male"   "female"
##  [65] "male"   "male"   "male"   "male"   "male"   "male"   "female" "male"
##  [73] "male"   "male"   "male"   "female" "male"   "female" "male"   "male"
##  [81] "male"   "female" "male"   "male"   "male"   "male"   "male"   "female"
##  [89] "female" "female" "male"   "female" "male"   "female" "female" "male"
##  [97] "male"   "female" "male"   "male"   "male"   "male"   "female" "female"
## [105] "male"   "male"   "male"   "male"   "male"   "male"   "female" "male"
## [113] "male"   "female" "female" "male"   "male"   "female" "male"   "male"
## [121] "male"   "female" "male"   "male"   "male"   "female" "female" "male"
## [129] "male"   "female" "male"   "male"   "male"   "male"   "female" "female"
## [137] "male"   "male"   "male"   "male"   "male"   "male"   "male"   "male"
## [145] "male"   "male"   "male"   "male"   "male"   "female" "male"   "female"
```

```
## [153] "female" "male"   "male"   "male"   "male"   "male"   "male"   "male"
## [161] "male"   "male"   "female" "female" "male"   "male"   "male"   "female"
## [169] "male"   "female" "male"   "male"   "female" "female" "male"   "male"
## [177] "male"   "male"   "male"   "male"   "male"   "female" "male"   "male"
## [185] "female" "female" "male"   "male"   "male"   "male"   "male"   "male"
## [193] "male"   "female" "female" "male"   "male"   "female" "female" "male"
## [201] "female" "female" "male"   "female" "male"   "male"   "male"   "male"
## [209] "male"   "female" "female" "male"   "male"   "female" "male"   "male"
## [217] "female" "female" "female" "male"   "female" "female" "female" "male"
## [225] "female" "female" "male"   "female" "male"   "male"   "female" "female"
## [233] "male"   "female" "female" "male"   "male"   "male"   "female" "male"
## [241] "male"   "female" "female" "male"   "female" "male"   "male"   "male"
## [249] "male"   "male"   "male"   "male"   "male"   "female" "male"   "female"
## [257] "female" "female" "male"   "male"   "female" "female" "female" "male"
## [265] "male"   "male"   "male"   "male"   "male"   "male"   "male"   "male"
## [273] "male"   "female" "male"   "male"   "female" "female" "male"   "female"
## [281] "male"   "male"   "female" "male"   "male"   "male"   "female" "male"
## [289] "male"   "male"   "male"   "female" "male"   "male"   "female" "male"
## [297] "male"   "female" "male"   "male"   "male"   "female" "male"
```

Subsetting our data based on a value for a column

```
clev[clev$ChestPain == "typical",]
```

```
##     Age    Sex ChestPain RestBP Chol   Fbs RestECG MaxHR ExAng Fluoroscopy
## 1    63   male   typical    145  233  TRUE       2   150    no           0
## 21   64   male   typical    110  211 FALSE       2   144   yes           0
## 22   58 female   typical    150  283  TRUE       2   162    no           0
## 28   66 female   typical    150  226 FALSE       0   114    no           0
## 31   69 female   typical    140  239 FALSE       0   151    no           2
## 42   40   male   typical    140  199 FALSE       0   178   yes           0
## 60   51   male   typical    125  213 FALSE       2   125   yes           1
## 102  34   male   typical    118  182 FALSE       2   174    no           0
## 113  52   male   typical    118  186 FALSE       2   190    no           0
## 125  65   male   typical    138  282  TRUE       2   174    no           1
## 142  59   male   typical    170  288 FALSE       2   159    no           0
## 151  52   male   typical    152  298  TRUE       0   178    no           0
## 183  42   male   typical    148  244 FALSE       2   178    no           2
## 184  59   male   typical    178  270 FALSE       2   145    no           0
## 197  69   male   typical    160  234  TRUE       2   131    no           1
## 200  59   male   typical    160  273 FALSE       2   125    no           0
## 212  38   male   typical    120  231 FALSE       0   182   yes           0
## 216  56   male   typical    120  193 FALSE       2   162    no           0
## 244  61   male   typical    134  234 FALSE       0   145    no           2
## 263  60 female   typical    150  240 FALSE       0   171    no           0
## 275  59   male   typical    134  204 FALSE       0   162    no           2
## 276  64   male   typical    170  227 FALSE       2   155    no           0
## 299  45   male   typical    110  264 FALSE       0   132    no           0
##            Thal AHD
## 1         fixed  No
## 21       normal  No
## 22       normal  No
## 28       normal  No
## 31       normal  No
## 42  reversable  No
```

```
## 60      normal  No
## 102     normal  No
## 113      fixed  No
## 125     normal Yes
## 142 reversable Yes
## 151 reversable  No
## 183     normal  No
## 184 reversable  No
## 197     normal  No
## 200     normal Yes
## 212 reversable Yes
## 216 reversable  No
## 244     normal Yes
## 263     normal  No
## 275     normal Yes
## 276 reversable  No
## 299 reversable Yes
```

Additional comparisons

You can also use ">=", "!=", ">", "<", and "<=":
"!=" means "not equal"
We had 23 individuals with "typical" chest pain, how many have something else?

```
sum(clev$ChestPain != "typical")
```

```
## [1] 280
```

23 + 280 = 303, which is correct! The number of individuals with and without "typical" chest pain adds up to our total.

Basic functions

sum()

In addition to adding up numbers (e.g. in a column),
sum() will treat TRUE as 1 and FALSE as 0.
This can be used to quickly tally the number of TRUE for a column:

```
sum(clev$Fbs)
```

```
## [1] 45
```

or condition:

```
sum(clev$ChestPain == "typical")
```

```
## [1] 23
```

length()

length() can also tell us how many values there are:

```
length(clev$ChestPain)
```

```
## [1] 303
```

mean()

```
mean(clev$Age)
```

```
## [1] 54.43894
```

Mean will treat also TRUE as 1 and FALSE as 0 and give us a proportion:

```
mean(clev$ChestPain == "typical")
```

```
## [1] 0.07590759
```

median()

```
median(clev$Age)
```

```
## [1] 56
```

sd() (standard deviation)

```
sd(clev$Age)
```

```
## [1] 9.038662
```

IQR() (interquartile range)

```
IQR(clev$Age)
```

```
## [1] 13
```

max() (maximum value)

```
max(clev$Age)
```

```
## [1] 77
```

Dealing with "NA"

If we try to sum the column for Fluoroscopy, we see it runs into an issue:

```
sum(clev$Fluoroscopy)
```

```
## [1] NA
```

is.na()

Some data is missing... how many rows are affected?

```
is.na(clev$Fluoroscopy)
```

```
##   [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [25] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [37] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [49] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [61] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

```
##  [73] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [85] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [97] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [109] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [121] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [133] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [145] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [157] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE
## [169] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [181] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [193]  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [205] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [217] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [229] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [241] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [253] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [265] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [277] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE
## [289] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [301] FALSE FALSE  TRUE
```

Adding sum():

```
sum(is.na(clev$Fluoroscopy))
```

```
## [1] 4
```

which()

which() tells us which rows return TRUE for a condition. We can combine it with is.na():

```
which(is.na(clev$Fluoroscopy))
```

```
## [1] 167 193 288 303
```

4 is not a worrying number. We can add "na.rm = T" to sum() to drop the individuals with missing values:

```
sum(clev$Fluoroscopy, na.rm = T)
```

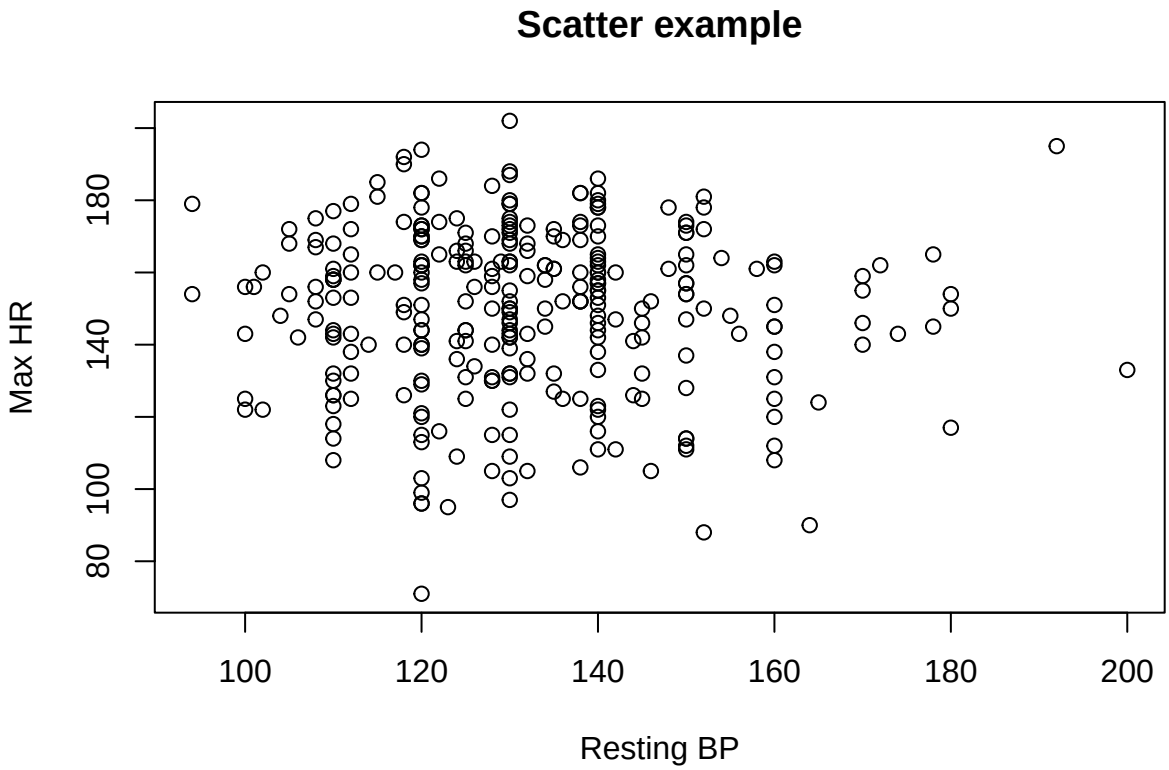
```
## [1] 201
```

Plotting

Scatter plot

Basic scatter plot needs x values and y values, e.g.: plot(valuesforxaxis, valuesforyaxis, xlab = "This is the x label", ylab = "This is the y label", main = "This is the plot title")

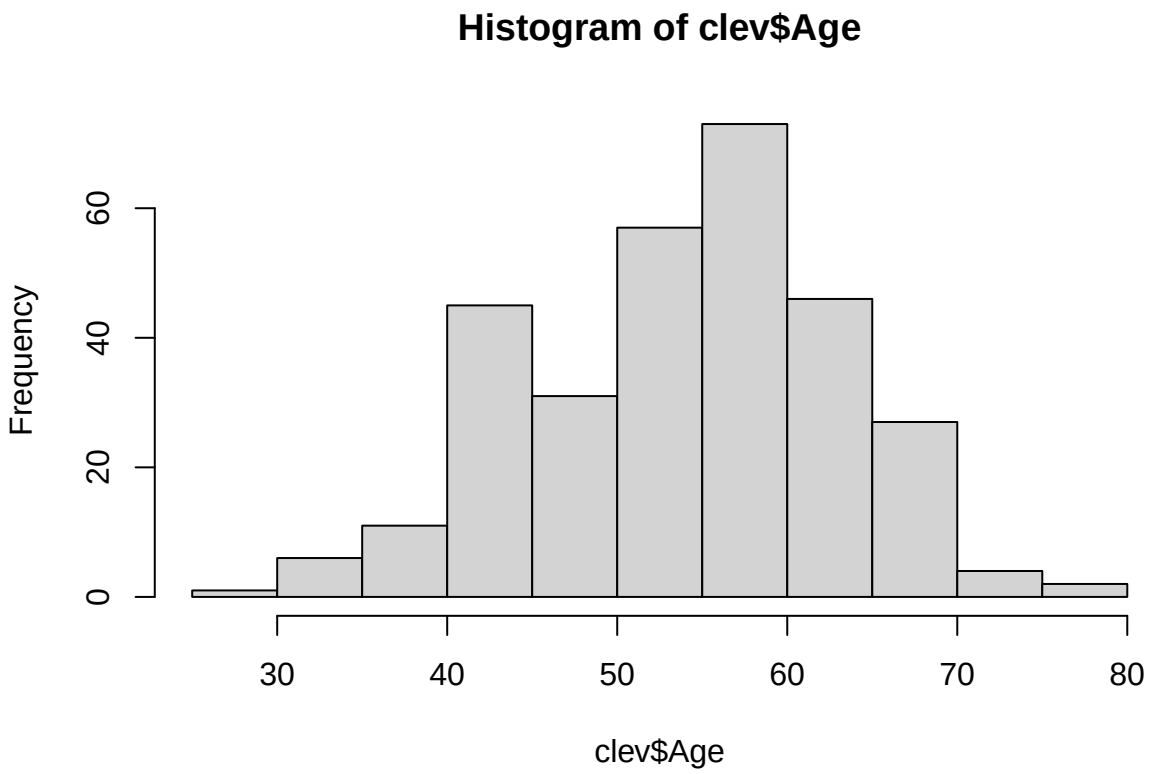
```r
plot(clev$RestBP, clev$MaxHR, xlab="Resting BP", ylab="Max HR", main="Scatter example")
```



Histogram

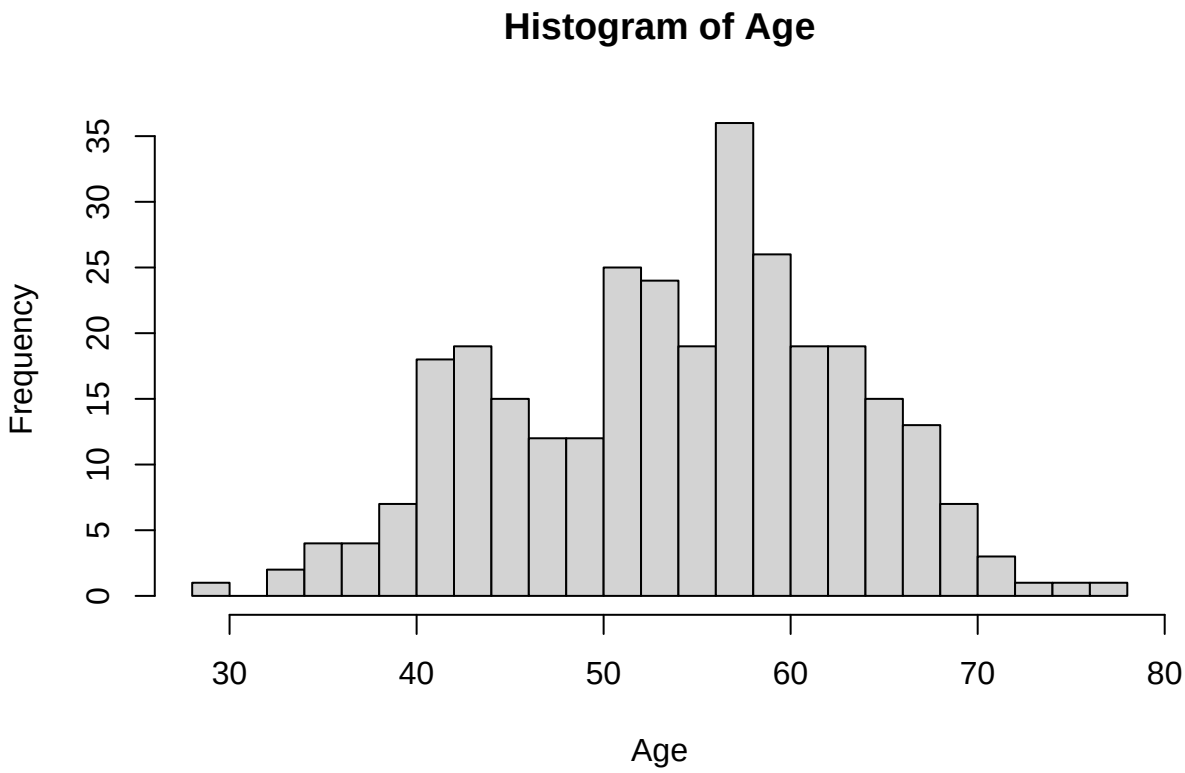
hist()

```r
hist(clev$Age)
```



Setting the number of bins in a histogram (and x label/title too):

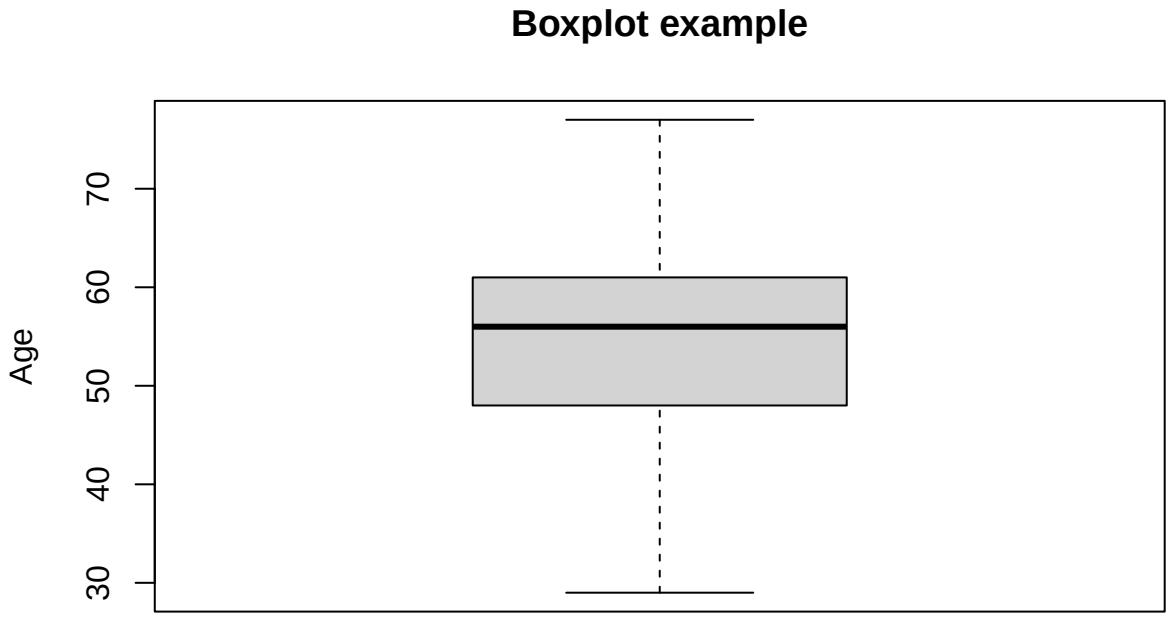
```
hist(clev$Age, breaks=20, xlab="Age", main="Histogram of Age")
```



Box plot

boxplot()

```
boxplot(clev$Age, ylab="Age", main="Boxplot example")
```



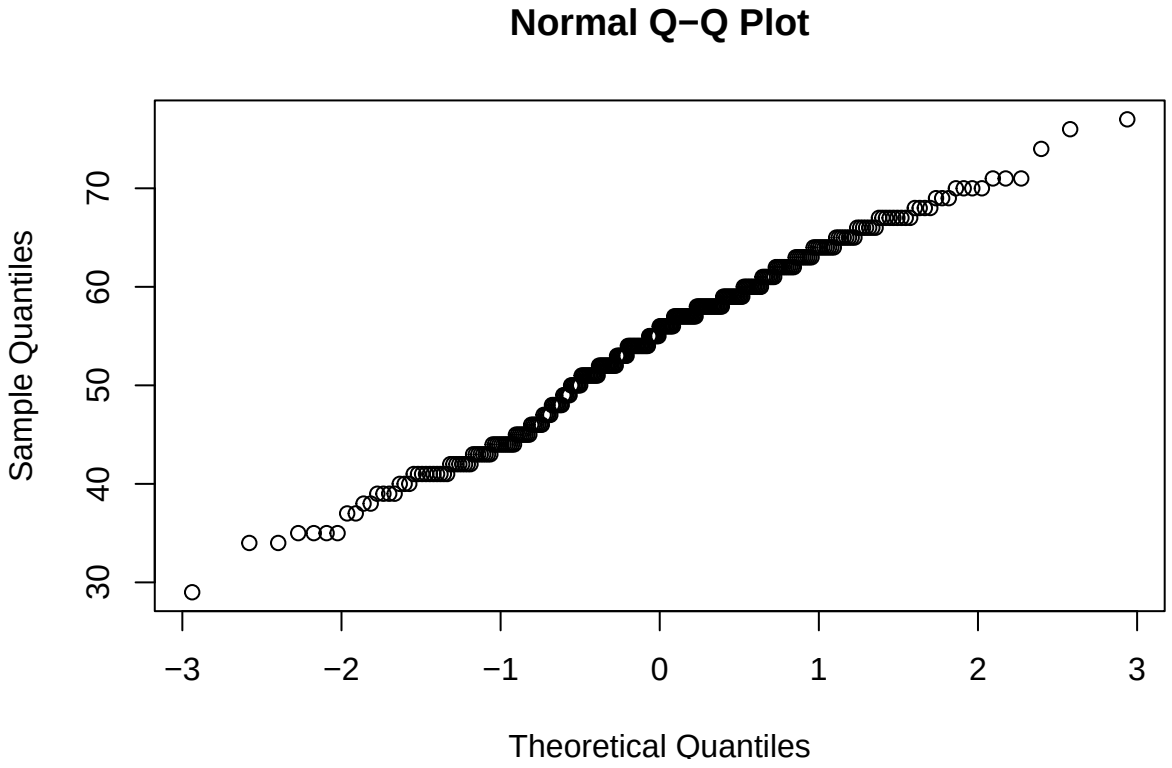
We can separate plots based on another variable (side-by-side plot)

```
boxplot(clev$RestBP~clev$Sex, xlab="Sex", ylab="Resting BP")
```



##QQ plots qqnorm()

```
qqnorm(clev$Age)
```




```
clev$Sex == "male"
```

```
##   [1]  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE FALSE  TRUE  TRUE  TRUE FALSE
##  [13]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
##  [25]  TRUE FALSE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [37]  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE  TRUE  TRUE
##  [49] FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [61] FALSE FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE
##  [73]  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE
##  [85]  TRUE  TRUE  TRUE FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE FALSE  TRUE
##  [97]  TRUE FALSE  TRUE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE
## [109]  TRUE  TRUE FALSE  TRUE  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## [121]  TRUE FALSE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## [133]  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [145]  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE FALSE  TRUE  TRUE  TRUE
## [157]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE FALSE
## [169]  TRUE FALSE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [181]  TRUE FALSE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [193]  TRUE FALSE FALSE  TRUE  TRUE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE
## [205]  TRUE  TRUE  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## [217] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE
## [229]  TRUE  TRUE FALSE FALSE  TRUE FALSE FALSE  TRUE  TRUE  TRUE FALSE  TRUE
## [241]  TRUE FALSE FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [253]  TRUE FALSE  TRUE FALSE FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE  TRUE
## [265]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE
## [277] FALSE FALSE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE  TRUE
## [289]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## [301]  TRUE FALSE  TRUE
```

Remember that "mean()" will treat TRUE as 1 and FALSE as 0:

```
mean(clev$Sex == "male")
```

```
## [1] 0.679868
```

More examples

Individuals under a certain age:

```
clev$Age <= 65
```

```
##   [1]  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [13]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [25]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [37]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [49]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [61]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE
##  [73]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE
##  [85]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
##  [97]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE
## [109]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [121]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [133]  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [145]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE FALSE
## [157]  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [169]  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [181]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE  TRUE
## [193]  TRUE  TRUE FALSE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [205]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE
## [217]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE
## [229]  TRUE FALSE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [241]  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [253]  TRUE  TRUE  TRUE  TRUE FALSE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
## [265]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE  TRUE
## [277] FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [289]  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE
## [301]  TRUE  TRUE  TRUE
```

Total number of individuals < 65:

```
sum(clev$Age <= 65)
```

```
## [1] 270
```

Proportion of individuals < 65:

```
mean(clev$Age <= 65)
```

```
## [1] 0.8910891
```

We can save this subset as a variable:

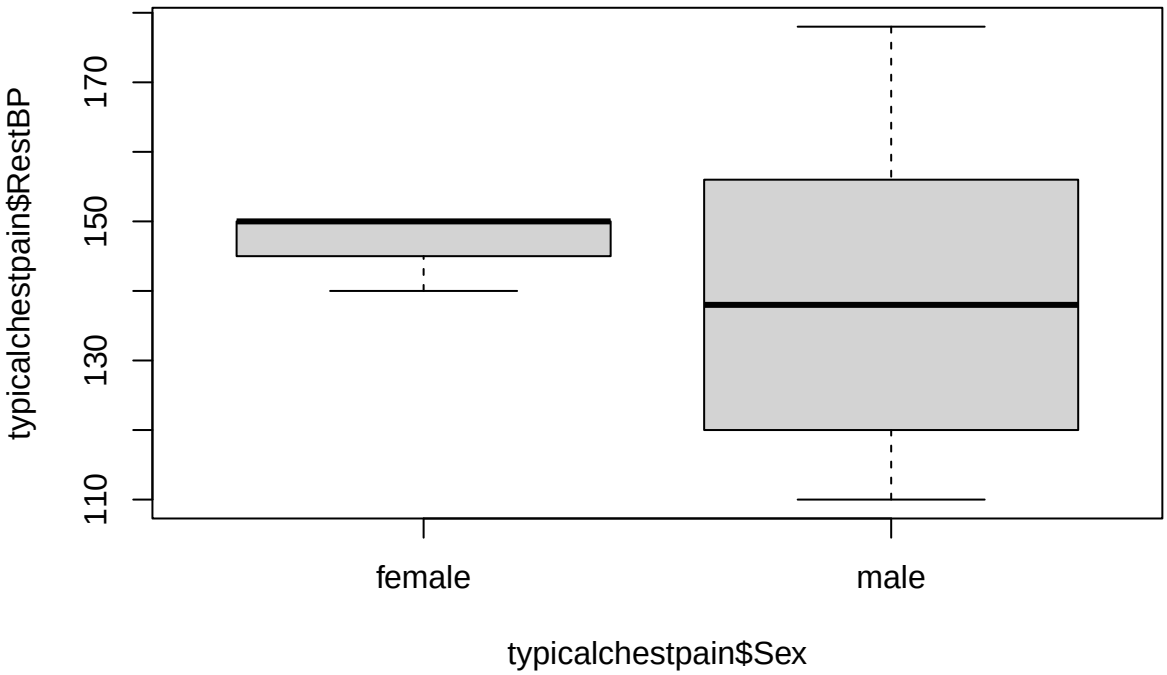
```
typicalchestpain <- clev[clev$ChestPain == "typical",]
typicalchestpain
```

```
##     Age    Sex ChestPain RestBP Chol   Fbs RestECG MaxHR ExAng Fluoroscopy
## 1    63   male   typical    145  233  TRUE       2   150    no           0
## 21   64   male   typical    110  211 FALSE       2   144   yes           0
## 22   58 female   typical    150  283  TRUE       2   162    no           0
```

```
## 28   66 female   typical    150  226 FALSE       0   114    no           0
## 31   69 female   typical    140  239 FALSE       0   151    no           2
## 42   40   male   typical    140  199 FALSE       0   178   yes           0
## 60   51   male   typical    125  213 FALSE       2   125   yes           1
## 102  34   male   typical    118  182 FALSE       2   174    no           0
## 113  52   male   typical    118  186 FALSE       2   190    no           0
## 125  65   male   typical    138  282  TRUE       2   174    no           1
## 142  59   male   typical    170  288 FALSE       2   159    no           0
## 151  52   male   typical    152  298  TRUE       0   178    no           0
## 183  42   male   typical    148  244 FALSE       2   178    no           2
## 184  59   male   typical    178  270 FALSE       2   145    no           0
## 197  69   male   typical    160  234  TRUE       2   131    no           1
## 200  59   male   typical    160  273 FALSE       2   125    no           0
## 212  38   male   typical    120  231 FALSE       0   182   yes           0
## 216  56   male   typical    120  193 FALSE       2   162    no           0
## 244  61   male   typical    134  234 FALSE       0   145    no           2
## 263  60 female   typical    150  240 FALSE       0   171    no           0
## 275  59   male   typical    134  204 FALSE       0   162    no           2
## 276  64   male   typical    170  227 FALSE       2   155    no           0
## 299  45   male   typical    110  264 FALSE       0   132    no           0
##           Thal AHD
## 1        fixed  No
## 21      normal  No
## 22      normal  No
## 28      normal  No
## 31      normal  No
## 42  reversable  No
## 60      normal  No
## 102     normal  No
## 113      fixed  No
## 125     normal Yes
## 142 reversable Yes
## 151 reversable  No
## 183     normal  No
## 184 reversable  No
## 197     normal  No
## 200     normal Yes
## 212 reversable Yes
## 216 reversable  No
## 244     normal Yes
## 263     normal  No
## 275     normal Yes
## 276 reversable  No
## 299 reversable Yes
```

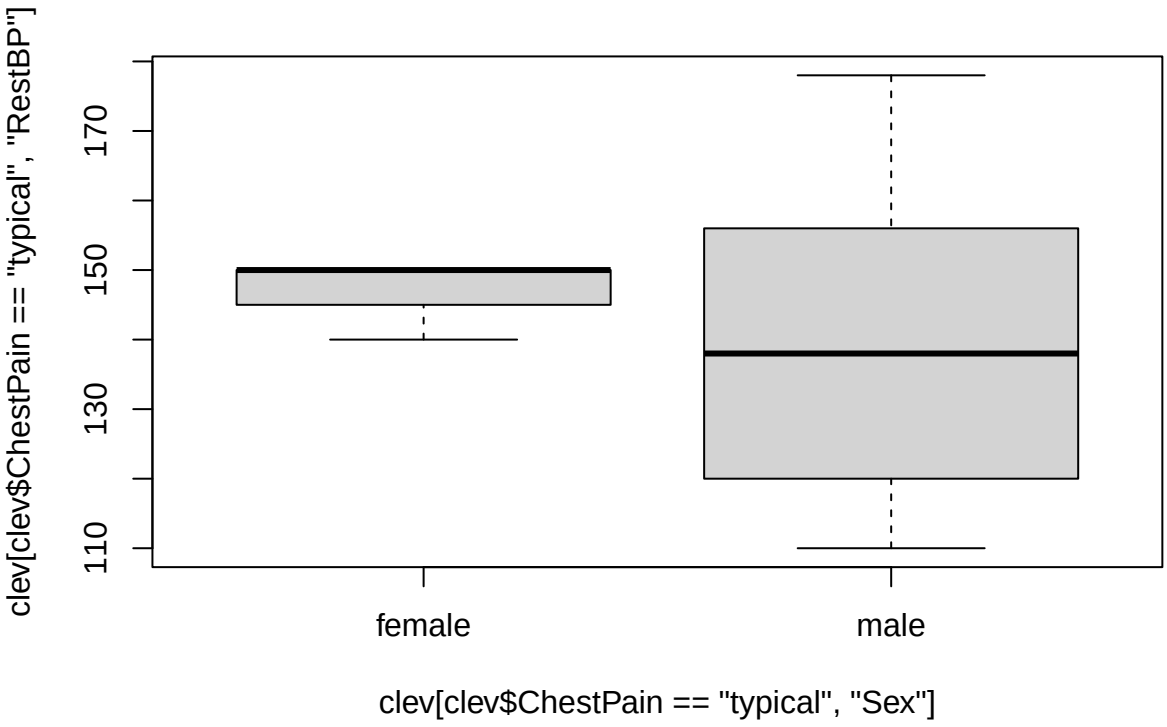
We can plot using this new variable:

```
boxplot(typicalchestpain$RestBP~typicalchestpain$Sex)
```



And it will give the same output as subsetting from the original data:

```
boxplot(clev[clev$ChestPain=="typical","RestBP"]~clev[clev$ChestPain=="typical","Sex"])
```



Knitting/preview

If any code block had an error above, the output will run into an issue. For some students, the pdf option may not work or require additional installation. Using HTML or Word and saving to pdf is also acceptable.